

Structural Integrity and Defect Rates in Pre-Filled Glass Syringes

ABSTRACT: Root cause analysis of micro-fractures in pre-filled syringe barrels under thermal shock.

1. Background

Packaging quality complaints (PQC) account for 15% of recalls.

Batch PF-4401 showed elevated flange stress.

2. Quality Evaluation

Defect rate: 0.03% in high-humidity storage.

Recommend updated annealing protocol.